



Fig. 2: An example of a set-top box hardware

age like hard disks or flash drives.

Digital storage. Digital storage is another optional component inside a set-top box. It is needed for persistent storage of any kind of data including audio/video. It communicates with the processor and is controlled by STB software modules and storage drivers. In fact, it is a hard disk which acts as a digital storage media inside an STB.

CA module. The conditional access (CA) module can be considered as the most important component of a set-top box. This module is virtually the lifeline of all the leading set-top box solution providers. This peripheral, called Integrated Conditional Access Module (ICAM), is placed before the demultiplexer to be used for descrambling the encrypted signal, and also provide a smartcard interface for various security features.

Return path. This is an optional component. A return path is used by an STB to communicate back with the HeadEnd and send data packets. The return path can be present in various forms like a PSTN line connection, a cable modem in case of cable STBs or an ethernet jack in IP set-top box using ADSL broadband modems. For example, a return path becomes a necessary component if a user needs

to purchase PPV (pay per view) events directly from the set-top box using credit cards.

Other peripherals. There are other optional peripherals present inside an STB like RS-232 port, USB port, Wi-Fi enablers, Bluetooth devices. These peripherals are used inside an STB for enhanced interactivity with the consumer and provide better user experience. Basically, these components or peripherals can be used for creating advanced features like file sharing and plugging in external devices.

Flow of data

As you must have understood the roles of various hardware modules inside an STB, let's move on to the next level, where we'll go through the flow of data between these hardware while we tune an STB to a particular channel.

The tuner receives modulated digital transmission from the antenna and passes it on to the demodulator. This demodulator takes into account the type of demodulation (like QPSK) and forward error correction to give out a transport stream, which is a digital stream of bytes known as data packets. According to DVB standards, a transport stream is of the size of 188 bytes. It then goes into the demux where the content is separated

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